

Yadi Cao

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RESEARCH FOCUS

AI4Science, AI for fusion, numerical methods, computational fluid dynamics, and computational mechanics

BIO

Yadi Cao develops novel **AI methods to scale up engineering research by reducing experimental costs and labor of experts**. His research has made a significant impact in fusion energy and computational fluid dynamics: **TGLF-WINN**, which delivers a **45x computational speedup** for fusion simulations and is **integrated into General Atomics' tokamak pipeline**, and **BSMS-GNN**, which enables learning realistic **turbomachinery simulations on million-node meshes**.

Yadi is a postdoctoral researcher at UCSD with Professor Rose Yu. He completed his Ph.D. at UCLA, co-advised by Professors Chenfanfu Jiang and Demetri Terzopoulos. He is seeking a tenure-track faculty position and is open to collaboration and mentoring opportunities.

EDUCATION

University of California, Los Angeles (UCLA)

Los Angeles, CA

Ph.D. in Computer Science

2021–2024

- Thesis: Advancing physics-based simulations: Integrating conventional and machine learning approaches for enhanced computational efficiency
- Advisors: Chenfanfu Jiang (Applied Math) and Demetri Terzopoulos (Computer Science)

University of British Columbia (UBC)

Kelowna, BC, Canada

MASc in Mechanical Engineering

2016–2018

- Thesis: Analytical and numerical study of plug flow inside round/concentric microchannels
- Advisor: Sunny Ri Li (Mechanical Engineering)

University of Science and Technology of China (USTC)

Hefei, Anhui, China

BEng

2012–2016

PUBLICATIONS [[GOOGLE SCHOLAR](#)] [TOTAL CITATIONS: 537, H-INDEX: 10]

Journal Articles

- **Cao, Y***, Liu, Y* (equal contribution), Yang, L, Yu, R, Schaeffer, H, Osher, S. VICON: Vision In-Context Operator Networks for Multi-Physics Fluid Dynamics Prediction. TMLR 2026.
- **Cao, Y***, Zhang, F*, Liu, W* (equal contribution), Neiser, T, Meneghini, O, Smith, S, Nazikian, R, Sammulu, B, Yu, R. TGLF-WINN: Data-Efficient Deep Learning Surrogate for Turbulent Transport Modeling in Fusion. NeurIPS ML4PS workshop. Nuclear Fusion (IF: 4.0), 2026.
- Somasekharan, N, Yue, L, **Cao, Y**, Li, W, Emami, P, Bhargav, P S, Acharya, A, Xie, X, Pan, S. CFD-LLMBench: A Benchmark Suite for Evaluating Large Language Models in Computational Fluid Dynamics. DMLR 2026.
- Yang, J, Bhat, M, Hu, B, **Cao, Y**, Dehmamy, N, Walters, R, Yu, R. Discovering Symbolic Differential Equations with Symmetry Invariants. TMLR 2026.
- Wang, P, **Cao, Y**, Russell, C, Shen, Y, Luo, J, Zhang, M, Heng, S, Luo, X. Delta: Dual Consistency Delving with Topological Uncertainty for Active Graph Domain Adaptation. TMLR 2025.
- **Cao, Y**, Zhao, Y, Li, M, Yang, Y, Choo, J, Terzopoulos, D, Jiang, C. Unstructured Moving Least Squares Material Point Methods: A Stable Kernel Approach with Continuous Gradient Reconstruction on General Unstructured Tessellations. Computational Mechanics (IF: 4.1), 2024.
- Fang, Y, Li, M, **Cao, Y**, Li, X, Wolper, J, Yang, Y, Jiang, C. Augmented Incremental Potential Contact for Sticky Interactions. IEEE TVCG (IF: 6.5), 2023.
- **Cao, Y**, Chen, Y, Li, M, Yang, Y, Zhang, X, Aanjaneya, M, Jiang, C. An Efficient B-Spline Lagrangian/Eulerian Method for Compressible Flow, Shock Waves, and Fracturing Solids. ACM TOG (SIGGRAPH, IF: 7.8), 2022.

- **Cao, Y**, Gao, X, Li, R. A Liquid Plug Moving in an Annular Pipe: Heat Transfer Analysis. International Journal of Heat and Mass Transfer (IF: 6.4), 2019.
- **Cao, Y**, Li, R. A Liquid Plug Moving in an Annular Pipe: Flow Analysis. Physics of Fluids (IF: 4.3), 2018.
Editor's Pick.

Conference Proceedings

- **Cao, Y***, Lai, S*, Huang, J*, Zhang, Y*, Lawrence, Z*, Bhakta, R*, Thomas, I F*, Cao, M* (equal contribution), Tsai, C, Zhou, Z, Zhao, Y, Liu, H, Marinoni, A, Arefiev, A, Yu, R. SimulCost: A Cost-Aware Benchmark for Automating Physics Simulations with LLMs. ICML 2026.
- Lu, H, **Cao, Y**, Yi, W, Abari, O. mmDiff: A Noise-Robust Differentiable Ray-Tracing Framework for mmWave Scene Calibration and Channel Prediction. CVPR 2026 Findings.
- Lyu, B*, **Cao, Y*** (equal contribution), Watson-Parris, D, Bergen, L, Berg-Kirkpatrick, T, Yu, R. Adapting While Learning: Grounding LLMs for Scientific Problems with Intelligent Tool Usage Adaptation. ICML 2025.
- Maheshwari, S, Mohanty, A, **Cao, Y**, Razu, S, McCulloch, A, Yu, R. BIGE: Biomechanics-informed GenAI for Exercise Science. L4DC 2025.
- Sun, F, Huang, Z, **Cao, Y**, Luo, X, Wang, W, Sun, Y. DoMiNO: Down-scaling Molecular Dynamics with Neural Graph Ordinary Differential Equations. ICLR 2025 Workshop on Machine Learning Multiscale Processes, 2025.
- Huang, Z, Zhao, W, Gao, J, Hu, Z, Luo, X, **Cao, Y**, Chen, Y, Sun, Y, Wang, W. Physics-Informed Regularization for Domain-Agnostic Dynamical System Modeling. NeurIPS 2024. **Best Paper Award**, NeurIPS DLDE workshop 2023.
- **Cao, Y**, Chai, M, Li, M, Jiang, C. Efficient Learning of Mesh-Based Physical Simulation with Bi-Stride Multi-Scale Graph Neural Network. ICML 2023.
- Li, X, **Cao, Y**, Li, M, Yang, Y, Schroeder, C, Jiang, C. PlasticityNet: Learning to Simulate Metal, Sand, and Snow for Optimization Time Integration. **Spotlight**, NeurIPS 2022.

Under Review Journal Articles

- Yue, L, Somasekharan, N, Zhang, T, **Cao, Y**, Pan, S. Foam-Agent 2.0: An End-to-End Composable Multi-Agent Framework for Automating CFD Simulation in OpenFOAM. ArXiv: 2509.18178. NeurIPS ML4PS workshop. In submission to CMAME (IF: 7.5).
- Ding, Y, Wu, K, **Cao, Y**, Magdon-Ismael, M, Gao, J. Computing Equilibria in Complex Systems with Billions of Interactions. In submission to PNAS NEXUS (IF: 3.8).

Preprints

- Wang, H, **Cao, Y**, Huang, Z, Liu, Y, Hu, P, Luo, X, Song, Z, Zhao, W, Liu, J, Sun, J, Zhang, S. Recent Advances on Machine Learning for Computational Fluid Dynamics: A Survey. ArXiv: 2408.12171.
- Zhang, Y, **Cao, Y**, Sun, S, Yu, R. CAED-Agent: An Agentic Framework to Automate Simulation-Based Experimental Design.

ACADEMIC AND TECHNICAL TALKS

Building the Cost-Aware Scientific Agent

CSE-259 AI Seminar, UCSD (Host: Prithviraj Ammanabrolu)

Oct, 2025

Efficient Learning of Multi-Physics Simulation by Vision In-Context Operator Learning

CMU KAAI Workshop (Host: Matt Ho)

Sep, 2026

SIAM UQ26 Mini-Symposium on In-context learning for PDEs and inverse problems (Host: Fei Lu)

Mar, 2026

Yale Lu Group Reading Group (Host: Lu Lu)

Aug, 2025

Algorithms for Multiphysics Models in the Post-Moore's Law Era, LANL (Host: Brian O'Shea)

Jun, 2025

Physics-Guided AI and Its Applications to Fusion

IEEE International Conference on Plasma Science Symposium (Host: Youngsoo Choi)

Jun, 2026

Lyncean Group of San Diego (Host: Bill Hagan)

Jan, 2026

Lawrence Livermore National Laboratory (Host: Ben Zhu)

Dec, 2025

CAST (Chinese Association for Science & Technology) (Host: Yongfeng Wu)

Aug, 2025

UCSD MAE+CER Winter 2025 Energy Seminar, UCSD (Host: George Tynan)

Feb, 2025

National Artificial Intelligence Research Resource Virtual Session Panelists (Host: Nitin Sukhija) Oct, 2024

Advancing Physics-based Simulation with Unstructured Discretization and Machine Learning

UCLA Level Set Lab Reading Group (Host: Stanley Osher) Feb, 2024

UCSD Rose STL Lab Reading Group (Host: Rose Yu) Jan, 2024

Yale Lu Group Reading Group (Host: Lu Lu) Jan, 2024

Caltech Anima AI + Science Lab Reading Group (Host: Anima Anandkumar) Jan, 2024

INDUSTRY EXPERIENCE

Google Student Researcher 2023–2024

Google Los Angeles, CA

Research Intern 2022

Snap Los Angeles, CA

SDE Intern 2021

Taichi Graphics Beijing, China

CAE/FEM Software Research Developer 2019–2020

shonCloud Tech/shonDynamics Suzhou, Jiangsu, China

TEACHING

Teaching Fellow at University of California, Los Angeles

Operating Systems Principles (CS 111) (Instructors: Paul Eggert, Peter Reiher) 6 Quarters, 2022–2024

Introduction to Algorithms and Complexity (CS 180) (Instructor: Matthew Ferland) Summer 2023

Teaching Assistant at University of British Columbia

Heat Transfer Applications (ENGR385) (Instructor: Ri Li) Spring 2018

Measurement Principles in Thermal Fluids (ENGR479) (Instructor: Ri Li) Winter 2018

Matter and Energy (APSC182) (Instructors: Shelir Ebrahimi, John Brereton) 2 Semesters, 2016–2017

MENTORING

Current Mentees

• **Undergraduate Students:**

- Lucas Hlaing (UCSD)
- Zach Lawrence (UCSD; next stop: PhD student at UW–Madison)

Alumni

• **Undergraduate Students:**

- Chloe Huang (University of Michigan; next stop: PhD student at Rose Lab)
- Wesley Liu (UCSD; next stop: Master's student at USC)
- Yang Zhang (Peking University; next stop: PhD student at UIUC)
- Leo Lai (CUHK-Shenzhen; next stop: Master's student at HKUST)
- Bohan Lyu (Tsinghua University; next stop: visiting student at Princeton University)
- Darin Djapri (UCSD; next stop: continuing study at UCSD)
- Yuwei Ren (UCSD; next stop: continuing study at UCSD)

• **Master's Students:**

- Futian Zhang (UCSD; next stop: Software Engineer at ServiceNow)
- Shubh Maheshwari (UCSD)
- Howard Tsai (UCSD; next stop: Software Engineer at TSMC)

Programs

Gear Up2Research (GEAR) 2025

UCSD San Diego, CA

Early Research Scholars Program (ERSP) 2024–2025

UCSD San Diego, CA

Summer Training Academy for Research Success (STARS) 2024–2025

UCSD San Diego, CA

AWARDS AND HONORS

UCLA Non-residential Tuition Grant	2022–2024
UCLA Graduate Fellowship	2021
UBC Graduate Fellowship	2017–2018
NSERC Funding for Exchange Research	2017
USTC Alumni Fellowship of B.C. Canada	2017
Bronze Prize, Zhongwei-cup Energy & Environment Protection Contest	2015
Silver Prize, National Outstanding Undergraduate Scholarship	2013–2014

RESEARCH COMMUNITY SERVICE

Area Chair

- ICML, AI for Science Workshop (2026)
- NeurIPS, AI for Science Workshop (2025)

Program Committee

- AAAI (2025, 2026)
- IJCAI (2024)

Reviewer (excluding committee services listed above)

- CVPR (2026)
- ECCV (2026)
- L4DC (2026)
- ICLR (2024-2026)
- ICML (2024-2026)
- NeurIPS (2023, 2025-2026)
- JMLR (Journal of Machine Learning Research) (2025)
- TMLR (Transactions on Machine Learning Research) (2025-2026)
- ACM SIGGRAPH (2024-2025)
- IEEE Transactions on Visualization and Computer Graphics (2025-2026)
- ICLR Workshop: Tackling Climate Change with Machine Learning (2025)
- Processes (MDPI) (2025-2026)
- Pacific Graphics (2022, 2024)
- Physics of Fluids (2018-2024)